

Taiabur Rahman

Machine Learning Engineer | Medical Imaging (AI) | Fullstack Developer

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Profile — ML Engineer and Medical Imaging specialist with **4+ years** of fullstack experience in healthcare SaaS and an **Erasmus Mundus Master** in Medical Imaging & Applications (France–Italy–Spain). Research background at **INSERM Dijon** on deep learning–based 3D brain segmentation. Proficient in end-to-end pipelines: data preprocessing, model training (PyTorch, TensorFlow), REST API deployment (FastAPI, Docker, AWS) and frontend integration (Vue.js, React). Eager to contribute to R&D or product teams in French medical-tech or AI sectors.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Sept. 2025 – Present

BLACKBIRD.AI — New York, USA (Remote)

- Developing and improving ML models for large-scale data analysis and AI system performance
- Contributing to AI architecture, data pipelines, and monitoring in an agile team
- Collaborating across engineering and research teams on model deployment

Graduate Research Intern — Deep Learning & Neuroscience

Feb. 2024 – Aug. 2024

INSERM UMR 1231, NEUROGEMM — Dijon, France

- Built an automated **nnU-Net** segmentation pipeline for high-resolution 3D histological mouse brain volumes (Python, PyTorch)
- Preprocessed and managed large-scale 3D imaging datasets on GPU cluster
- Tuned and evaluated model performance; validated results with neuroscientists
- Contributed to scientific documentation and internal research presentations

Full-Stack Developer (Full-time)

Dec. 2017 – Aug. 2022

Vision Eye Hospital — Dhaka, Bangladesh

- Designed and maintained a hospital ERP / HIMS solution deployed across **5 hospitals**
- Built modules for prescriptions, diagnostics, surgical workflows and patient records
- Stack: **Laravel, Vue.js, MySQL**; managed DevOps with Nginx/Apache on Linux

RESEARCH PROJECTS

3D Mouse Brain Segmentation — nnU-Net, PyTorch [\[github\]](#)

Automated pipeline for high-res histological 3D brain data; trained on GPU cluster.

CT Image Registration — Elastix + SimpleITK [\[github\]](#)

Rigid and affine registration framework for multi-modal medical images.

3D Brain Tissue Segmentation (MRI) — TensorFlow, PyTorch, U-Net [\[github\]](#)

WM / GM / CSF tissue delineation from volumetric MRI scans.

Skin Lesion Classification — VGG16, ResNet50, Keras [\[github\]](#)

Transfer learning on dermoscopic images with cross-validation fine-tuning.

Alzheimer's Classification — Multi-modal — SVM, Scikit-learn [\[github\]](#)

Fused MRI features + gene expression data with ensemble methods.

Retinal Image Analysis (2D) — Classical CV pipeline [\[github\]](#)

Thresholding + morphological operations for hard exudate segmentation.

SOFTWARE PRODUCTS

AI-MIQA — Medical Image Quality Control Platform

ai-miqa.com — AI-powered automated quality control and analysis for medical

EDUCATION

Erasmus Mundus Joint Master

Medical Imaging & Applications (MAIA)

Sept. 2022 – Aug. 2024

Université de Bourgogne — Dijon, **France**

Università degli Studi di Cassino — Italy

Universitat de Girona — Spain

Medical image segmentation & registration, deep learning, computer-aided diagnosis, surgical robotics, e-health.

MSc — Software Engineering

Tianjin University, China

BSc — Computer Science & Engineering

Bangladesh University, Bangladesh

TECHNICAL SKILLS

ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, nnU-Net, SimpleITK
AI / LLM	OpenAI API, Gemini API, LangChain
Medical Img.	Segmentation, Registration, CAD, 3D volume analysis
Backend	Python (FastAPI, Flask, Django), PHP (Laravel)
Frontend	Vue.js, React, TypeScript, JavaScript, jQuery
DevOps / Cloud	AWS (EC2, S3), Docker, CI/CD, Linux, Nginx
Databases	PostgreSQL, MySQL, SQLite
Tools	Git, REST API, Jupyter, Apache

LANGUAGES

Bengali — Native

English — Upper-intermediate (B2)

French — Beginner (A1–A2, actively learning)

TRAINING & SCHOOLS

17th EXCITE Summer School

Biomedical Imaging

ETH Zurich & Univ. of Zurich, Switzerland

Sept. 2023

Winter School 2023

Università di Cassino, Italy

Feb. 2023

TARGET ROLES (FRANCE)

images; enhances diagnostic accuracy for research and clinical settings.

Hospital Information Management System (HIMS)

Web-based ERP active in 5 hospitals (Bangladesh). Stack: Laravel, Vue.js, MySQL.

- ML / AI Engineer — Medical Imaging, Health-Tech
- Fullstack Software Engineer (Python / Vue / React)
- R&D Engineer — Computer Vision, Image Analysis
- DevOps / MLOps (Docker, AWS, CI/CD)

ATS KEYWORDS

Medical Imaging · 3D Segmentation · Deep Learning · PyTorch · nnU-Net · SimpleITK · Computer Vision · Python · FastAPI · Vue.js · React · Docker · AWS · CI/CD · REST API · SaaS · INSERM · Erasmus Mundus · MLOps

Paris, France — Open to on-site, hybrid or remote opportunities in France and Europe